Chengpeng Wang

Postdoctoral Fellow in Computer Science, Purdue

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RESEARCH INTEREST

Software Engineering, Machine Learning, and Programming Languages, with a focus on combining symbolic static analysis and large language models to advance code auditing and strengthen software security and performance, particularly in the context of AI-assisted coding.

EDUCATION

Hong Kong University of Science and Technology *Aug 2019 - Dec 2023*
Ph.D. in Computer Science and Engineering
Committees: *Charles Zhang (advisor)*, *Xiangyu Zhang*, *Shing-Chi Cheung*, *Shuai Wang*
Dissertation: *Enhancing Reliability and Performance of Data-Centric Systems with Static Analysis*

Tsinghua University *Aug 2012 - Jul 2019*
B.S and M.S. in Software Engineering

ACADEMIC AND INDUSTRY APPOINTMENTS

Now Mar 2024	**Postdoctoral Fellow, West Lafayette, IN, USA** LLM-driven code auditing, working with Prof. Xiangyu Zhang
Aug 2022 Feb 2021	**Research Intern at Ant Group, Shenzhen, China** Performance optimization of database-backed application

SELECTED ACHIEVEMENTS

Open Source Contributions (Over 1,000 GitHub Stars) *By Sep. 2025*
Released LLM-driven code auditing tools and benchmarks, garnering 1,086 GitHub stars in total.

Discovered 1,000+ Zero-Day Bugs in Open-Source Projects *By Aug. 2025*
Discovered over 1,005 zero-day bugs affecting software security and performance, including 200 in the Linux kernel. More than 700 have been confirmed and patched by developers.

Industrial Impact of Static Analyzers *2021–2025*
Adopted by Ant Group, Microsoft, and Uber, uncovering hundreds of previously unknown vulnerabilities in production systems. Integrated into Amazon Trusted AI Challenge red-teaming solution, winning the $250k prize, and is now being commercialized as the flagship product of BiScope.AI.

Invited Industry Talks *2025*
Invited to deliver talks at the GitHub CodeQL Technical Forum, the DARPA Resilient Software Systems Demo Day, and the RSA Conference 2025.

HONORS AND AWARDS

NeurIPS 2025 Spotlight	*2025*
Postdoc Travel Award, Purdue	*2024*
ACM SIGARCH Best Paper Award, ASPLOS'24	*2024*
UGC Research Travel Grant, HKUST	*2022, 2023*
Ant Group Outstanding Collaboration Award	*2023*
ACM SIGPLAN Distinguished Paper Award, OOPSLA'22	*2022*
ACM SIGPLAN PAC Award	*2022*
Outstanding Tutor Award, Student Learning and Development Center, THU	*2019*
Future Scholar Award, honored 100 students at Tsinghua University	*2016*
Scholarship for Academic Excellence, Tsinghua University	*2013, 2014*

GRANTS

PI. "*Neuro-Symbolic Vulnerability Detection for Reliable Software Systems*". Submitted to ***ARC Discovery Early Career Researcher Award Scheme, Australia***.

PI. "*Synthesizing Realistic Benchmarks for LLM-Aided Code Security Auditing*". Researcher Access Program, Funded by ***OpenAI***

Co-Advisor, with Xiangyu Zhang (PI). "*Amazon Trusted AI Challenge: An Agentic Red Teaming Framework for Code LLMs*". Funded by ***Amazon*** (acceptance rate: 11%, 10/90).

PUBLICATIONS

Highlight: I authored 7 first-author papers and 2 co-first-author papers (labeled with #) published in top-tier conferences and journals, including **ICSE**, **FSE**, **ASE**, **TOSEM**, **OOPSLA**, **ECOOP**, **ICML**, **NeurIPS**, and **EMNLP**. The underlined first authors are Ph.D. students under my mentoring.

PAPERS IN PROGRESS

[P1] Jinyao Guo, **Chengpeng Wang**, Dominic Deluca, Jinjie Liu, Zhuo Zhang, Xiangyu Zhang, BugScope: Learn to Find Bugs Like Humans. (Under Review)

[P2] Yifei Gao, **Chengpeng Wang**, Pengxiang Huang, Xuwei Liu, Mingwei Zheng, and Xiangyu Zhang, PR^2: Peephole Raw Pointer Rewriting with LLMs for Translating C to Safer Rust. (Under Review)

[P3] **Chengpeng Wang**, Yifei Gao, Wuqi Zhang, Xuwei Liu, Jinyao Guo, Mingwei Zheng, Qingkai Shi, and Xiangyu Zhang, NESA: Relational Neuro-Symbolic Static Program Analysis. (Under Review)

JOURNAL PUBLICATIONS

[J1] Chao Wang, Li Lin, **Chengpeng Wang**, Jiafeng Huang, Congxia Wu, and Rongxin Wu, ReachCheck: Compositional Library-Aware Call Graph Reachability Analysis in the IDEs, In **TOSEM 2025**: Transactions on Software Engineering and Methodology

[J2] Hao Ling, Heqing Huang, **Chengpeng Wang**, Yuandao Cai, and Charles Zhang, GiantSan: Efficient Operation-Level Memory Sanitization with Segment Folding, In **TOCS 2024**: The ACM Transactions on Computer Systems. (Invited extension of [C11])

[J3] Wensheng Tang, Dejun Dong, Shijie Li, **Chengpeng Wang**, Peisen Yao, Jinguo Zhou, and Charles Zhang, Octopus: Scaling Value-Flow Analysis via Parallel Collection of Realizable Path Conditions, In **TOSEM 2023**: Transactions on Software Engineering and Methodology

[J4] **Chengpeng Wang**, Wenyang Wang, Peisen Yao, Qingkai Shi, Jinguo Zhou, Xiao Xiao, and and Charles Zhang, Anchor: Fast and Precise Value-Flow Analysis for Containers via Memory Orientation, In **TOSEM 2022**: Transactions on Software Engineering and Methodology

CONFERENCE PUBLICATIONS

[C1] Mingwei Zheng, **Chengpeng Wang**, Xuwei Liu, Jinyao Guo, Shiwei Feng, and Xiangyu Zhang, RFCAudit: AI Agent for Auditing Protocol Implementations Against RFC Specifications, In **ASE 2025**: International Conference on Automated Software Engineering.

[C2] Danning Xie, Mingwei Zheng, Xuwei Liu, Jiannan Wang, **Chengpeng Wang**, Lin Tan, and Xiangyu Zhang, CORE: Benchmarking LLMs' Code Reasoning Capabilities through Static Analysis Tasks, In **NeurIPS 2025** (Spotlight): Annual Conference on Neural Information Processing Systems

[C3] Jinyao Guo[#], **Chengpeng Wang**[#], Xiangzhe Xu, Zian Su, and Xiangyu Zhang. RepoAudit: An Autonomous LLM-Agent for Repository-Level Code Auditing. In **ICML 2025**: International Conference on Machine Learning.

[C4] Yuan Li, Peisen Yao, Kan Yu, **Chengpeng Wang**, Yaoyang Ye, Song Li, Meng Luo, Yepang Liu, and Kui Ren. Understanding Industry Perspectives of Static Application Security Testing Evaluation, In **FSE 2025**: International Conference on the Foundations of Software Engineering.

[C5] Mingwei Zheng, Danning Xie, Qingkai Shi, **Chengpeng Wang**, and Xiangyu Zhang, Validating Network Protocol Parsers with Traceable RFC Document Interpretation, In **ISSTA 2025**: International Symposium on Software Testing and Analysis.

[C6] **Chengpeng Wang**, Wuqi Zhang, Zian Su, Xiangzhe Xu, Xiaoheng Xie, and Xiangyu Zhang, LLMDFA: Analyzing Dataflow in Code with Large Language Models, In **NeurIPS 2024**: Annual Conference on Neural Information Processing Systems

[C7] **Chengpeng Wang**, Wuqi Zhang, Zian Su, Xiangzhe Xu, and Xiangyu Zhang, Sanitizing Large Language Models in Bug Detection with Data-Flow, In **Findings of EMNLP 2024**: Empirical Methods in Natural Language Processing

[C8] Li Lin, Zongyin Hao, **Chengpeng Wang**, Zhuangda Wang, Rongxin Wu, and Gang Fan, SQLess: Dialect-Agnostic SQL Query Simplification, In **ISSTA 2024**: International Symposium on Software Testing and Analysis

[C9] **Chengpeng Wang**, Jipeng Zhang, Rongxin Wu, and Charles Zhang, DAInfer: Inferring API Aliasing Specifications from Library Documentation via Neurosymbolic Optimization, In **FSE 2024**: International Conference on the Foundations of Software Engineering

[C10] Bowen Zhang, Wei Chen, Peisen Yao, **Chengpeng Wang**, Wensheng Tang, and Charles Zhang, SIRO: Empowering Version Compatibility in Intermediate Representations via Program Synthesis, In **ASPLOS 2024**: ACM Conference on Architectural Support for Programming Languages and Operating Systems

[C11] Hao Ling, Heqing Huang, **Chengpeng Wang**, Yuandao Cai, and Charles Zhang, GiantSan: Efficient Memory Sanitization with Segment Folding, In **ASPLOS 2024**: ACM Conference on Architectural Support for Programming Languages and Operating Systems 🏆 **ACM SIGARCH Best Paper Award**

[C12] Rongxin Wu, Yuxuan He, Jiafeng Huang, **Chengpeng Wang**, Wensheng Tang, Qingkai Shi, Xiao Xiao, and Charles Zhang, LibAlchemy: A Two-Layer Persistent Summary Design for Taming Third-Party Libraries in Static Bug-Finding Systems, In **ICSE 2024**: International Conference on Software Engineering

[C13] **Chengpeng Wang**, Peisen Yao, Wensheng Tang, Gang Fan, Charles Zhang, Synthesizing Conjunctive Queries for Code Search, In **ECOOP 2023**: European Conference on Object-Oriented Programming

[C14] Zongyin Hao, Quanfeng Huang, **Chengpeng Wang**, Jianfeng Wang, Yushan Zhang, Rongxin Wu, and Charles Zhang, Pinolo: Detecting Logical Bugs in Database Management Systems with Approximate Query Synthesis, In **ATC 2023**: USENIX Annual Technical Conference

[C15] **Chengpeng Wang**, Gang Fan, Peisen Yao, Fuxiong Pan, and Charles Zhang, Verifying Data Constraint Equivalence in FinTech Systems, In **ICSE 2023**: International Conference on Software Engineering

[C16] Wensheng Tang[#], **Chengpeng Wang**[#], Peisen Yao, Rongxin Wu, Xianjin Fu, Gang Fan, and Charles Zhang, DCLink: Bridging Data Constraint Changes and Implementations in FinTech Systems, In **ASE 2023** : International Conference on Automated Software Engineering

[C17] Rongxin Wu, Minglei Chen, **Chengpeng Wang**, Gang Fan, Jiguang Qiu, and Charles Zhang, Accelerating Build Dependency Error Detection via Virtual Build, In **ASE 2022** : International Conference on Automated Software Engineering

[C18] **Chengpeng Wang**, Peisen Yao, Wensheng Tang, Qingkai Shi, and Charles Zhang: Complexity-Guided Container Replacement Synthesis, In **OOPSLA 2022**: SIGPLAN Conference on Objected Oriented Programming, Systems, Languages and Applications 🏆 **ACM SIGPLAN Distinguished Paper Award**

[C19] Gang Fan, **Chengpeng Wang**, Rongxin Wu, Xiao Xiao, Qingkai Shi, and Charles Zhang: Escaping Dependency Hell: Finding Build Dependency Errors with the Unified Dependency Graph, In **ISSTA 2020**: International Symposium on Software Testing and Analysis

WORKSHOP and POSTER PUBLICATIONS

[W1] Ming Liang, Xiaoheng Xie, Gehao Zhang, Xunjin Zheng, Peng Di, Wei Jiang, Hongwei Chen, **Chengpeng Wang**, and Gang Fan, RepoGenix: Dual Context-Aided Repository-Level Code Completion with Language Models, In Poster Track, **ASE 2024**: International Conference on Automated Software Engineering

[W2] **Chengpeng Wang**, CodeSpider: Automatic Code Querying with Multi-modal Conjunctive Query Synthesis, In SRC track, **SPLASH 2022**: SIGPLAN conference on Systems, Programming, Languages, and Applications: Software for Humanity, Student Research Competition

[W3] **Chengpeng Wang**, Yixiao Yang, Han Liu, and Le Kang: Statistical API Completion Based on Code Relevance Mining, In **MAINT 2019**: International Workshop on Mining and Analyzing Interaction Histories

ACADEMIC SERVICES

Program Organizing Committee Member:

International Workshop on Language Models and Programming Languages (LMPL)	*2025*

Program Committee Member:

ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)	*2026*
International Conference on the Foundations of Software Engineering (FSE)	*2025, 2026*
International Symposium on Software Testing and Analysis (ISSTA)	*2025, 2026*
International Conference on Automated Software Engineering (ASE)	*2025*
International Conference on Automated Software Engineering (ASE), Industrial Track	*2024*
International Symposium on Software Reliability Engineering (ISSRE)	*2024, 2025*
Annual Conference on Neural Information Processing Systems (NeurIPS)	*2025*
International Conference on AI Foundation Models and Software Engineering (Forge)	*2024, 2025*
SPLASH, Student Research Contest Track	*2024*
International Workshop on Large Language Models for Code (LLM4Code)	*2026*

Journal Reviewer:

Transactions on Software Engineering and Methodology (TOSEM)	2025
Transactions on Software Engineering (TSE)	2024

Artifact Evaluation Committee Member:

International Conference on OOP, Systems, Languages, and Applications (OOPSLA)	*2024*
ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)	*2023*
International Conference on the Foundations of Software Engineering (FSE)	*2022*
International Symposium on Software Testing and Analysis (ISSTA)	*2022*

Volunteer:

International Conference on OOP, Systems, Languages, and Applications (OOPSLA)	*2022*
Student Volunteer at International Symposium on Software Testing and Analysis (ISSTA)	*2019*
Long-term mentor of SIGPLAN-M	*2024, 2025*

TEACHING

CS 510: Software Engineering, Purdue University	*Fall 2025*
CS 592: AI and Security, Purdue University	*Fall 2024*

COMP 3021: Java Programming, HKUST	*Spring/Fall 2022/2023*
COMP 4631: Computer and Communication Security, HKUST	*Fall 2021*
COMP 3111/H: Software Engineering, HKUST	*Fall 2020*
COMP 2011: Programming with C++, HKUST	*Spring 2020*
Haskell: Functional Language Programming, THU	*Spring 2019*
Automaton and Formal Logic, THU	*Fall 2018*
Drop-in Tutoring for STEM Courses, THU	*Spring/Fall 2018/2019*

ADVISING AND MENTORING

Jinyao Guo (Graduate Student, Purdue University) *Nov 2024 - Present*
Currently a Ph.D. student at Purdue University
Working on LLM-driven static analysis and coauthored ICML'25 [C3] and the preprint [P1]

Mingwei Zheng (Graduate Student, Purdue University) *Aug 2024 - Present*
Currently a Ph.D. student at Purdue University
Working on network protocol bug detection and coauthored ISSTA'25 [C5] and ASE'25 [C1].

Dominic DeLuca (Undergraduate Student, Purdue University) *May 2025 - Aug 2025*
Currently a third-year undergraduate student at Purdue University
Working on LLM-assisted program debugging and coauthored the preprint [P1]

Danning Xie (Graduate Student, Purdue University) *Oct 2024 - May 2025*
Currently a Ph.D. student at Purdue University
Working on evaluating LLMs in code reasoning tasks and coauthored NeurIPS'25 [C2].

Calix Barrus (Graduate Student, The University of Texas at San Antonio) *June 2025 - Aug 2025*
Currently a visiting Ph.D. student at Purdue University
Working on LLM-assisted call graph analysis for Java

Yifei Gao (Graduate Student, Purdue University) *May 2024 - April 2025*
Currently a Ph.D. student at Purdue University
Working on automated translation from C to safer Rust and coauthored the preprint [P2].

Pengxiang Huang (Graduate Student, Northwestern University) *May 2024 - Present*
Currently a Master student at Northwestern University
Working on automated translation from C to safer Rust and coauthored the preprint [P2].

Bowen Zhang (Graduate Student, HKUST) *Oct 2022 - Dec 2023*
Currently a Ph.D. student at Hong Kong University of Science and Technology
Worked on IR Translator Synthesis and coauthored ASPLOS'24 [C10].

PRESENTATIONS AND INVITED TALKS

RepoAudit: Human-like AI Auditor for Code Repositories

RSA Conference 2025, San Francisco	Apr 2025

Advances in AI-powered Code Security: Next-Level Bug Detection

Uber's Programming Systems Team. Virtual	Feb 2025
CodeQL@GitHub. Virtual	Jan 2025

Neuro-Symbolic Static Analysis for Reliable Software Systems

UMass Amherst	Sep 2025
University of British Columbia, Columbia University	Aug 2025
UC Davis, UIUC, UNSW	May 2025
AST Lab, ETH Zurich. Virtual	Mar 2025
Department of Computing, Hong Kong Polytechnic University	Jan 2025
Institute of Data Science, The University of Hong Kong	Dec 2024

Neuro-Symbolic Static Analysis for Reliable and Performant Software Systems
School of Computer Science, Nanjing University. Nanjing, China — Dec 2024

When Static Analysis Meets Large Language Models: A Neuro-Symbolic Approach
Ant Group. Virtual — Dec 2024
Boston Computation Club. Virtual — Nov 2024
AI-CyberSecurity Research Lunch Talk, Texas A&M University (TAMU). Virtual — Oct 2024
School of Software, Tsinghua University. Virtual — Oct 2024

Towards Enhancing Reliability and Performance of Data-Centric Systems with Static Analysis
School of Informatics, Xiamen University. Xiamen, China — Aug 2023

Synthesizing Conjunctive Queries for Code Search
ByteDance. Virtual — Jun 2023

Complexity-Guided Container Replacement Synthesis
AST Lab, ETH Zurich. Virtual — Mar 2023